Exam Date & Time: 30-Nov-2024 (02:30 PM - 05:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH END SEMESTER EXAMINATIONS, NOV/DEC 2024

SATELLITE COMMUNICATION [ECE 3126]

Marks: 50 **Duration: 180 mins.**

DESCRIPTIVE

Answer all the questions.

Instructions to Candidates: Answer ALL questions Missing data may be suitably assumed

1) With relevant diagrams and mathematical expressions, explain look angle determination (5)

A)

B) The apogee and perigee distances of a satellite orbiting in an elliptical orbit are respectively 45 000 km and 7000 km. Determine the following:

i. Semi-major axis of the elliptical orbit

ii. Orbit eccentricity (3)

iii. Distance between the center of the Earth and the center of the elliptical orbit

C) Discuss the differences between LEO and MEO satellites (2)

2) Find the maximum Doppler shift of the MEO satellite at an altitude

A) of 20,200 km when the satellite has an elevation angle of 10^0. Assume operating frequency=1575MHz and take $\mu = 39.8 \times 10^{13} Nm^2/kg$ (4)

B) With neat diagrams, explain the orbital effects in satellite communication (3)

C) With neat diagrams, explain the various causes for orbital perturbations and remedy for them (3)

3) A C-band earth station has an antenna with a transmit gain of 54 dB. The transmitter output power is set to 100 W at a frequency of 6.1 GHz. The signal is received by a satellite at a distance of 37,500 km by an antenna with a gain of 26 dB. The signal is then routed to a transponder with a noise temperature of 500 K, a bandwidth of 36 MHz, and a gain of 110 dB. (5)

A)

i. Calculate the path loss at 6.1 GHz.

ii. Calculate the power at the output port of the satellite antenna, in dBW.

iii. Calculate the noise power at the transponder input, in dBW, in a bandwidth of 36 MHz.

iv. Calculate the C/N ratio, in dB, in the transponder.

Calculate the carrier power, in dBW and in watts, at the transponder output.

B) A transmission line at temperature 290K is fed from a source whose noise temperature T_{in} =1450K. The input signal power S_i is 100picowatts(pW) and the signal bandwidth is 1GHz. The transmission line has a loss of 3dB. Calculate the $(SNR)_{in}$, the system noise temperature , the output signal power S_{out}, and the $(SNR)_{out}$ (3)

C) Satellite antenna makes 5^0 to cover India and surrounding ocean and neighbouring countries. Determine antenna diameter and gain at 4GHz (2)

4) With a neat diagram, discuss double conversion transponder used in communication subsystem of geosynchronous satellite (4)

A)

B) In a TDMA system having each burst preamble= 560 bits, reference burst size =576 bits, number of reference bursts=2, number of user bursts =12, Guard time equivalent number of bits =128bits, TDMA transmission rate =96.64 Mbps, Voice channel rate= 64kbps, Frame time = 2msec, find number of voice channel supported (3)

C) A 36 MHz transponder has maximum EIRP of 50dBw and has multiple carriers. Receive station G/T is -10dB/K, free space loss is 192 dB, miscellaneous loss is 6.6dB. Each 10kHz carrier needs 45dBHz C/No. Find number of carriers that can be supported by the transponder. (3)

5) With a neat block diagram explain the various subsystems of typical earth station used in geosynchronous satellite. (5)

A)

B) With a neat diagram explain the three segments of global positioning system (3)

C) With neat diagrams explain Cassegrain and offset fed Cassegrain antenna (2)

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